

Distributed real-time control architecture for electrohydraulic humanoid robots

Authors: Subhi Jleilaty, Anas Ammounah, Ghiath Abdulmalek (2023)

ABSTRACT

The authors propose a distributed real-time control architecture for humanoid robots, separating high-level cognitive tasks and trajectory planning from low-level joint actuators. Communication is managed via a dedicated real-time bus, ensuring deterministic step response.

RELEVANCE TO THE EB15 ROBOTIC ARM PROJECT

Supports the modular split of control responsibilities between the ESP32 supervisor and the Arduino Uno actuator nodes.

TECHNICAL ANALYSIS & SYSTEM INTEGRATION

This academic research reference documents crucial design paradigms directly relevant to the development of the EB15 six-degrees-of-freedom robotic manipulator. Under the distributed control framework designed for this thesis (featuring the ESP32 S3 as master and the Arduino Uno as slave), the mathematical formulations, communication latencies, and performance profiles analyzed by Subhi Jleilaty provide a benchmark for physical and algorithmic modeling. Specifically, this reference establishes solid validation criteria for timing constraints, closed-loop feedback via absolute magnetic encoders (AS5600), and vibration damping.